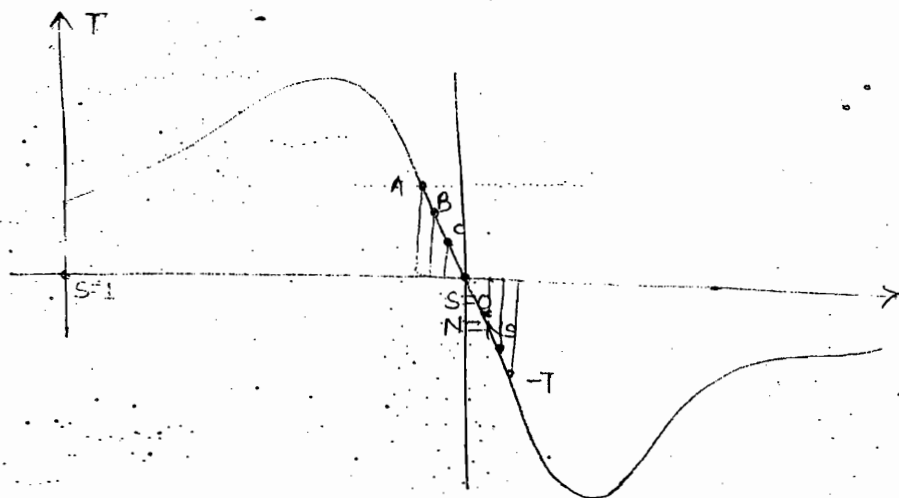
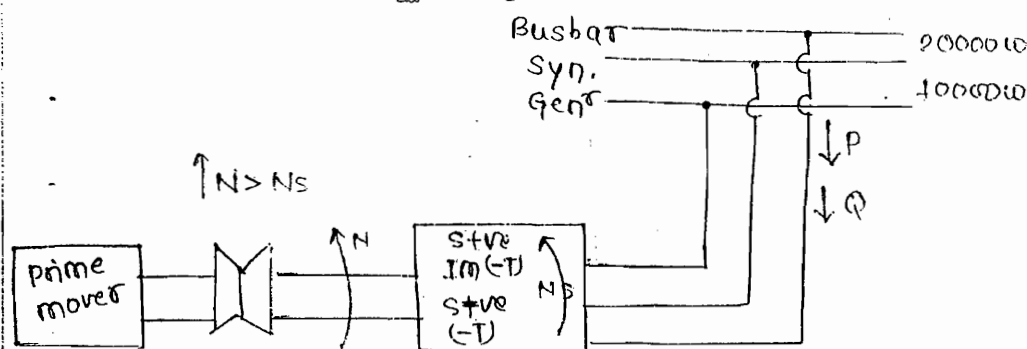


* Induction Generator →



- * If an IM is connected to suitable prime mover & rotated greater than its syn. speed it act as Indⁿ gen^r by supplying active power while drawing reactive power from bus (supply lines where it is connected)
- * It can't operate alone as it requires Q^* when it is connected to ∞ bus where syn. gen^r are operating across it the active power supplied to the bus freq. depends on the freq. of reactive power drawn from bus.
- Therefore in such condⁿ in its operating region the freq. of active power supplied is independent of speed variation.
- Therefore it is used for variable speed generation specially non-conventional like wind mills, mini hydro plants etc.

* It act as reactive power burden on existing syn. gen^r.

Therefore not preferred generally, but advantageous in specific power gen^m as the m/c is simple in design, less cost, maintenance free operation, no hunting, stability prob. & doesn't require any dc excitations.

* In such emergency applⁿ it can be used a load as self excited Indⁿ gen^r.

* Self excited indⁿ gen^r →

The principle is as similar to vol. build up in self excited shunt gen^r.

It requires some residual flux in rotor which is achieved through running m/c as IM for some time.

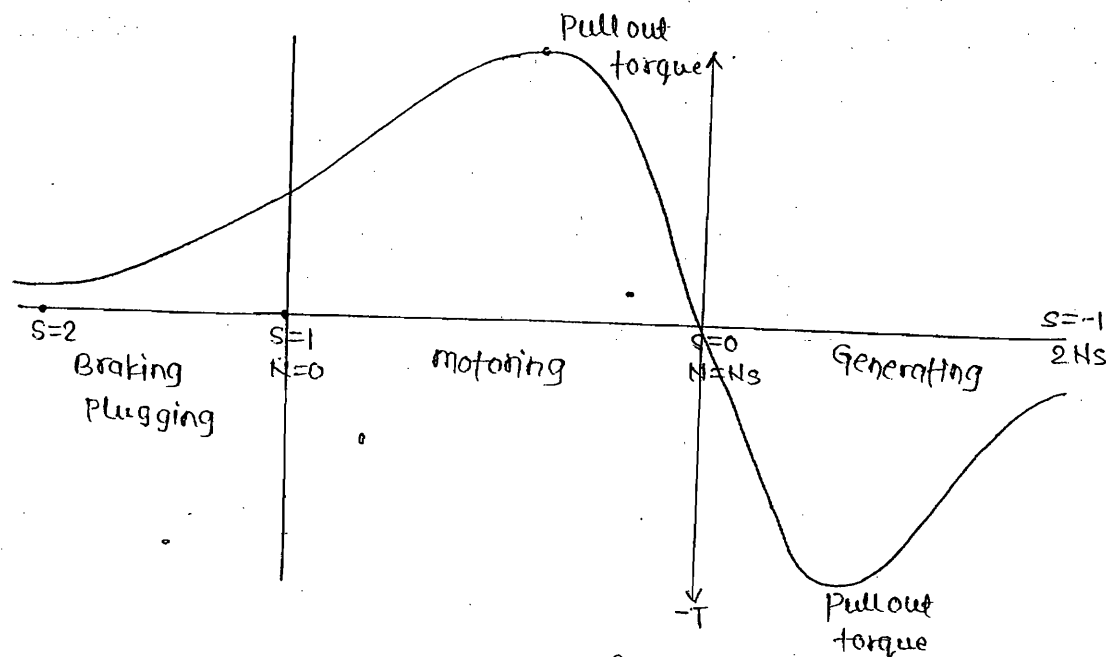
Once the rotor is rotated residual flux induce vol. in the stator where a capacitor bank is connected across it which supplies reactive power for flux development.

There is a cumulative build up of flux in air gap upto saturation of stator & rotor cores.

This is not suitable for loads which demand high reactive power eg. large IM while starting.

The capa. bank should supply the reactive power demand of gen^r as well as load.

Effective control is required to control reactive power otherwise bus vol. will vary.



* Special Constructions →

squirrel cage Im rotor → Generally around 90% motors are sq. cage type because of their own ad.

* But the basic limitation is due to their low starting torque.

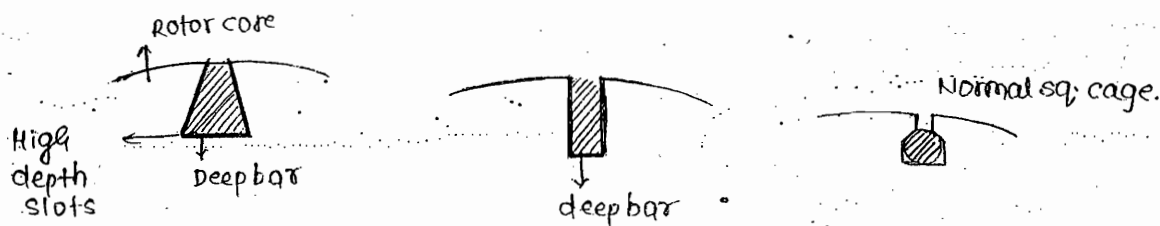
* In order to increase their starting torque w/o affecting their η must some special const. are adopted.

* The principle of cons is based on skin effect.

(1) Deep bar

(2) Double cage

* Deep bar →



* The rotor contains high depth slots where deep bars are placed & solidally sc. with 2 end rings.

* During starting rotor freq. is supply freq. skin effect dominates & the current goes to flow only on the top region of deep bar.

Therefore it offers high resistance to starting current which increases the starting torque while reducing starting current.

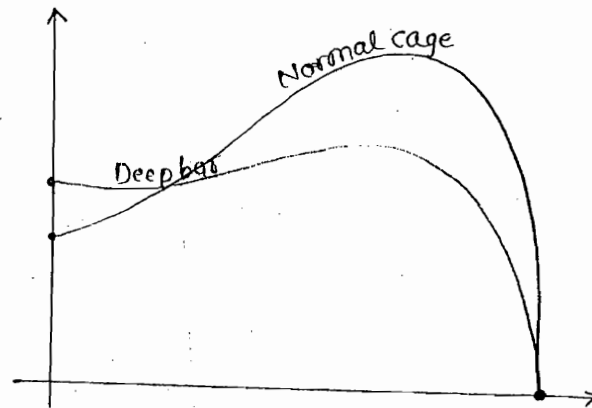
* When the motor runs the rotor freq. is slip freq. very low.

Therefore skin effect is negligible.

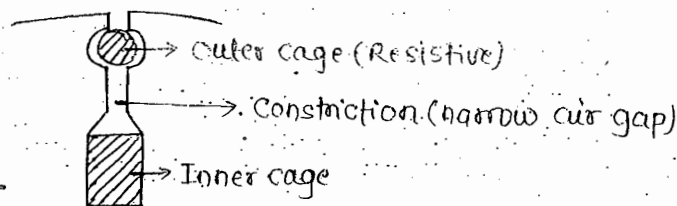
* Consequently the current shifts into bottom region which offers low resistance & running η are not affected.

* Due to deep slot (or) bar the leakage reactance at stand still is high.

Therefore this motor has less max^m (or) breakdown torque not suitable for high rating loads (or) doesn't have good overload capability used for small rating app comparatively which demand high starting torque.



* Double cage →



* It contains 2 cages :-

(1) Outer cage with small cross-sectional area high resistance.

(2) Inner cage is large cross-sectional area less resistance.

* Which are solidally so with end rings on both sides.

* During starting the freq. of rotor is supply freq. & starting current only flows in outer cage which offer high resistance to result in high starting torque while reducing starting current.

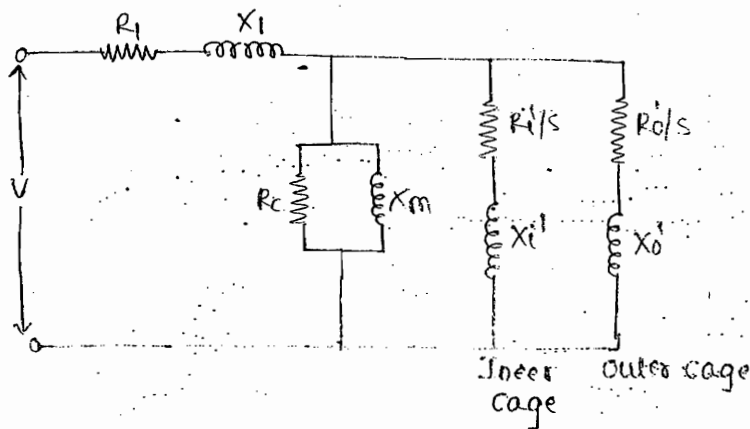
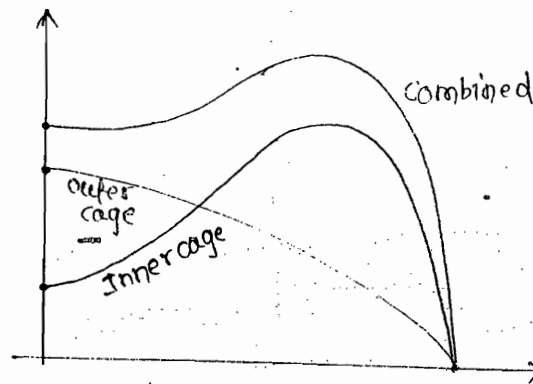
* Under running condⁿ rotor freq. will be negligible & skin effect is quite low.

Therefore the current shifts into inner cage which offer low resistance.

Starting torque increased w/o affecting running η much.

* The basic diff. b/w deep bar & double cage lies in constriction (narrow air gap b/w 2 cages).

* It reduce leakage reactance of rotor at stand still & result in more max^m or breakdown torque compare to deep bar.



* These special constructions are rarely preferred for specific app. specially less ratings compared to sq. cage but demand high starting torque.

* For large rating high starting torque requirements slip ring motors are preferred.

DATE-10/12/14

Crawling →

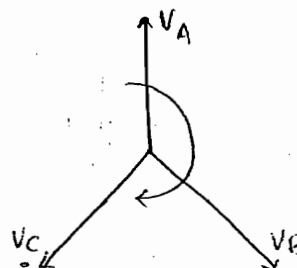
Odd	$3m$	3, 9,
	$3m-1$	5, 11,
	$3m+1$	7, 13,

Fundamental(f)

$$V_A = V_m \sin \omega t$$

$$V_B = V_m \sin(\omega t - 120^\circ)$$

$$V_C = V_m \sin(\omega t - 240^\circ)$$



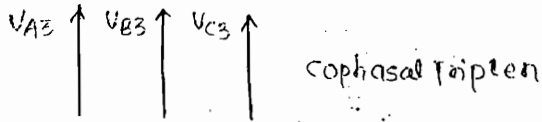
3rd harmonics (3f)

Sagar Sen
8871453536

$$V_{A3} = V_{m3} \sin 3\omega t$$

$$V_{B3} = V_{m3} \sin 3(\omega t - 120) = V_{m3} \sin 3\omega t$$

$$V_{C3} = V_{m3} \sin 3(\omega t - 240) = V_{m3} \sin 3\omega t$$



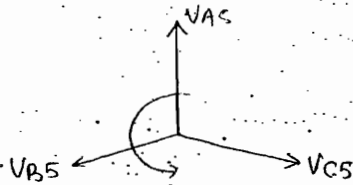
* They have no mutual ϕ displacement & called as cophasal which doesn't support 3 ϕ sys. & can't exist easily.

5th harmonics $\rightarrow$

$$V_{A5} = V_{m5} \sin 5\omega t$$

$$V_{B5} = V_{m5} \sin 5(\omega t - 120) = V_{m5} \sin (5\omega t - 240)$$

$$V_{C5} = V_{m5} \sin 5(\omega t - 240) = V_{m5} \sin (5\omega t - 120)$$



* They have mutual ϕ displacement 120 support 3 ϕ nature, exist easily but they have opposite ϕ seq. to that of fundamental.

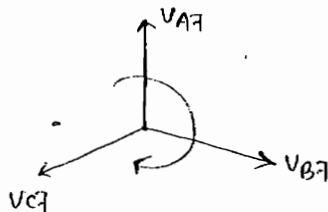
Therefore there affect will be in the braking region of IM ($s > 1$)

7th harmonics $\rightarrow$

$$V_{A7} = V_{m7} \sin 7\omega t$$

$$V_{B7} = V_{m7} \sin 7(\omega t - 120) = V_{m7} (7\omega t - 120)$$

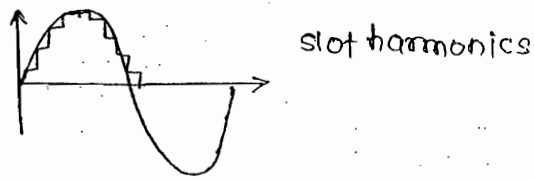
$$V_{C7} = V_{m7} \sin 7(\omega t - 240) = V_{m7} (7\omega t - 240)$$



* They are having mutual ϕ displacement of 120° exactly 3ϕ nature, easily exist & they are as identical to fundamental with same ϕ seq.

5th $\rightarrow$ Braking
7th $\rightarrow$ Motoring

* If harmonics applied in the supply they known as time harmonics.



* If a fundamental is produced by p no. of poles then fictitiously a 5th harmonic is produced by $5p$ & 7th is produced by $7p$ no. of poles.

$$N_{s7} = \frac{120f}{7P} = \frac{N_s}{7}$$

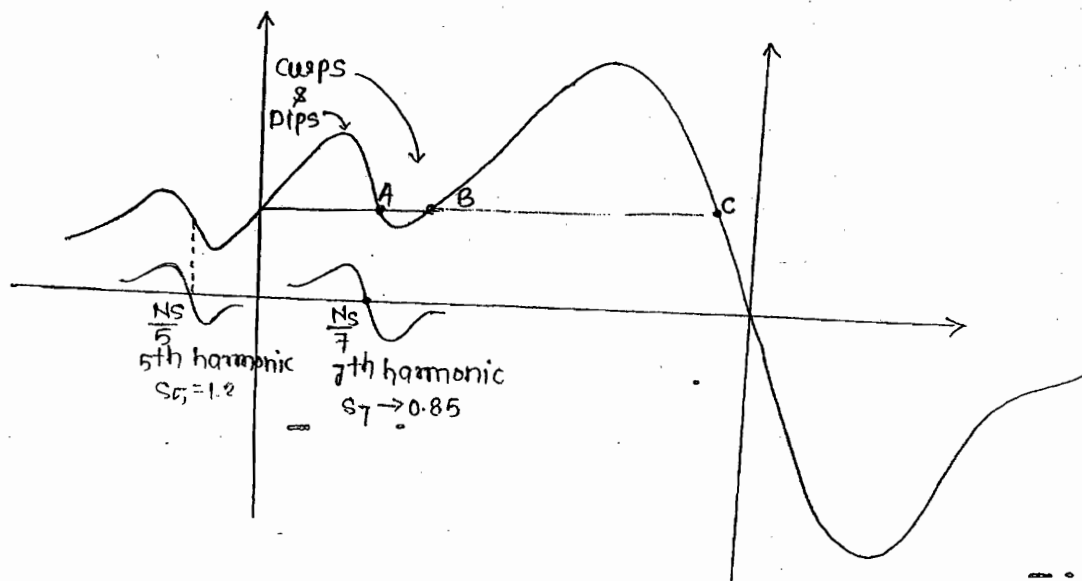
$$s_7 = \frac{N_s - \frac{N_s}{7}}{N_s} = \frac{6}{7}$$

$$N_{s5} = \frac{120f}{5P} = \frac{N_s}{5}$$

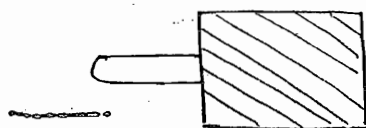
$$\therefore s_5 = \frac{N_s - \left(-\frac{N_s}{5}\right)}{N_s} = \frac{6}{5}$$

Because of
plugging

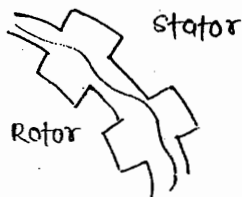
$$= 1.2$$



- * It is a different behaviour of sq. cage IM to run at reduced speed around $N_s/7$ during starting.
- * Due to time/slot harmonics the flux distribution will get distorted & contains harmonics in it.
- * The most effective harmonics is 7th in motoring region & 5th in the braking region. They distort fundamental T-S c/s in their operating regions & corresponding slips.
- * If the motor starts with low starting torque it may fall near $N_s/7$ & runs stably supplying a particular load torque even though (point A). Even though it is a stable point it is not desired operating pt. due to high slip, high current, huge noise, high loss, low speed.
- * If starting torque high eg. slip ring there is no effect of crawling even though it contains harmonic content inside.
- * In order to eliminate crawling effect due to harmonics stator wdg is uniformly distributed & short pitched.
- * Fractional slot wdg, semi closed slots are preferred.
- * The rotor slots are also skewed for better flux distribution & elimination of harmonic content.



- * With skewing it will also produce uniform torque smooth running with reduced noise (magnetic hum).



* Cogging →

- * It is a tendency of magnetic locking b/w stator & rotor teeth due to alignment b/n them.
- * If the rotor & stator slots (or) teeth are exactly equal & parallel to each other the alignment torque b/n them will dominate the accelerating torque, specially in sq cage motors as their starting torques are low (reduced Vol. starting).
- * Rotor refuse to start & being stationary with locking is known as cogging.
- * To eliminate this rotor slots are essentially skewed (inclined or twisted) skewing also increase the resistance of rotor bars & produce high starting torque comparatively.
- * But the running η is comparatively low.
- * Skewing also increased leakage reactance & reduce max^m torque comparatively.
- "While designing Im the slots of stator & rotor will made different & also doesn't contain common factor b/n them but the no. of poles should be same."

* Comparison b/n sq cage & slip ring Im →

- * Net air gap of sq cage is comparatively low & req. less magnetising current & operate at better PF.
- * Rotor resistance of solid bars in sq cage is comparatively low & the η is comparatively more.
- * The end ring doesn't contain overhang but in slip ring due to overhang & increased depth of slot leakage reactance is comparatively more.
- * max^m (or) breakdown torque & overload capacity is comparatively more in sq cage.
- * Construction is simple, low cost; mech. strong, less weight, maintenance free operation.

* Inverted IM (or) Rotor Fed $\rightarrow$

SRMF wrt stator at

* If an IM is supplied with 3 ϕ supply across its rotor by closing its stator it is called as IM rotor fed.

* It operates normally but rotor runs at opposite dirⁿ.

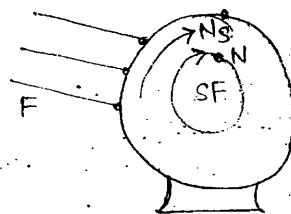
* It can be done only in slip ring type as the rotor is accessible through slip ring & brushes but conventionally motors are operated through stator.

SRMF wrt stator at N_S

RRMF wrt stator at $-N_S + N = N_S$

As rotor itself rotate at N in direction of mag. field

SRMF & RRMF wrt rotor: $N_S - N$
 $= S N_S$



* When rotor is fed with supply with freq. f , it will produce RRMF at speed N_S wrt rotor. But the stator induced emf, current produce torque & want to make stator rotate.

* As the stator is fixed part & rotor is designed for rotation inability to make the stator rotate will result in the rotor rotating in the opposite dirⁿ to that of magnetic field.

Rotor RMF wrt rotor at N_S

rotor RMF wrt stator at stationary point $N_S - N = S N_S$
(or) stationary point
in space

* Stator freq. is SF , therefore stator RMF wrt stator is at $S N_S$.

SRMF wrt stator at $S N_S$

SRMF & RRMF wrt rotor at $S N_S - (-N) \Rightarrow S N_S + N$
 N_S

$\therefore$ rotor rotate in opposite dirⁿ.

